Investigation: TAURUS ARMAS S.A. (CVM code 6173), fiscal year 2011, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Metalurgia e Siderurgia (origin: FCA); group: non-financial.
Trading codes (2023 FCA, version 1, filed on 2023-04-05): TASA3 (Ações Ordinárias, Bolsa, since 2011-07-08); TASA4 (Ações Preferenciais, Bolsa, since 2011-07-08).

Document used

2011 DFP, version 1, filed on 2012-03-27; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2012-03-27).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=15473

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −2.498 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 38 (n = 17)
- Percentile in group and year: 52 (n = 371)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 0.985 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 37 (n = 23)
- Percentile in group and year: 39 (n = 467)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.071 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 46 (n = 23)
- Percentile in group and year: 69 (n = 477)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.210 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 80 (n = 23)
- Percentile in group and year: 84 (n = 472)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 1.015 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 37 (n = 23)

Percentile in group and year: 27 (n = 478)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 0.960 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 21 (n = 17)
Percentile in group and year: 40 (n = 385)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.945 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 28 (n = 23)
Percentile in group and year: 33 (n = 471)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.005 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 67 (n = 23)
Percentile in group and year: 68 (n = 494)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 1.371 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 93 (n = 23)
Percentile in group and year: 89 (n = 486)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 4.8 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 35.1 days)
Percentile in sector and year: 57 (n = 22)
Percentile in group and year: 61 (n = 449)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 13.10 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 67 (n = 23)
Percentile in group and year: 70 (n = 438)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.039 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 41 (n = 23)
Percentile in group and year: 52 (n = 494)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 2.996 (index)
Components: X1 = 0.252; X2 = 0.096; X3 = 0.090; X4 = 0.406 (indices)
Percentile in sector and year: 48 (n = 24)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 2/9 (8 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 1
- Change in ROA (ΔROA): 0
- Cash above earnings (ACCRUAL): 0
- Change in leverage (ΔLEVER): 0
- Change in current ratio (ΔLIQUID): 0
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Highlighted in the following year's comparative: 3 (threshold of R$11.3 million, 1% of total assets of the version used). Values in R$ million:
DFC (indirect method) 6.01 Caixa Líquido Atividades Operacionais: 45.3 in the version used (v1, filed on 2012-03-27); 78.6 in the next DFP's comparative (v7, filed on 2014-03-28); difference of 33.3. Different value in the following year's comparative (restatement or reclassification).
BPA 1 Ativo Total: 1,126.7 in the version used (v1, filed on 2012-03-27); 1,114.3 in the next DFP's comparative (v7, filed on 2014-03-28); difference of −12.3. Different value in the following year's comparative (restatement or reclassification).
BPP 2.02 Passivo Não Circulante: 335.9 in the version used (v1, filed on 2012-03-27); 323.5 in the next DFP's comparative (v7, filed on 2014-03-28); difference of −12.3. Different value in the following year's comparative (restatement or reclassification).

Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 40 accounts above the R$1,000 tolerance (group a, totals and key lines: 3; group b, reclassifications: 37), in the annex at the end.

What was known on the date

- FCA 2011, v1: 2011-03-25
- DFP 2010, v1: 2011-03-30
- DFP 2010, v2: 2011-03-30
- DFP 2010, v3: 2011-04-12
- FCA 2011, v2: 2011-05-31
- FCA 2011, v3: 2011-08-12
- DFP 2011, v1: 2012-03-27
- DFP 2012, v1: 2013-03-26
- DFP 2012, v2: 2013-04-04
- DFP 2012, v3: 2013-04-24
- DFP 2012, v4: 2013-10-22
- DFP 2012, v5: 2013-10-24
- DFP 2012, v6: 2013-11-01
- DFP 2012, v7: 2014-03-28

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 2 failed.

- Cash flow closing balance = cash and cash equivalents on the balance sheet (PENÚLTIMO): difference above the tolerance of R$2,000
- Cash flow opening balance = cash and cash equivalents on the prior balance sheet (ÚLTIMO): difference above the tolerance of R$2,000

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
Excerpt checked literally: "Em nossa opinião, as demonstrações contábeis individuais referidas no primeiro parágrafo apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da Forjas Taurus S.A. em 31 de dezembro de 2011, o desempenho de suas operações e os seus fluxos de caixa para o exercício findo naquela data, de acordo com as práticas contábeis adotadas no Brasil."
Auditor in the 2021 FCA: Deloitte Touche Tohmatsu Auditores Independentes.
Auditor in the 2022 FCA: Deloitte Touche Tohmatsu Auditores Independentes.
Auditor in the 2023 FCA: Deloitte Touche Tohmatsu Auditores Independentes.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2012: KPMG Auditores Independentes -> Ernst & Young Auditores Independentes S.S. (transition: two firms in the 2012 FCA). The outgoing firm audited 8 consecutive fiscal years (2004 to 2011), by the start and end dates in the FCA.
- 2015: Ernst & Young Auditores Independentes S.S. -> KPMG Auditores Independentes. The outgoing firm audited 3 consecutive fiscal years (2012 to 2014), by the start and end dates in the FCA.
- 2020: KPMG Auditores Independentes -> Deloitte Touche Tohmatsu Auditores Independentes. The outgoing firm audited 5 consecutive fiscal years (2015 to 2019), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
- 2023: Deloitte Touche Tohmatsu Auditores Independentes -> Deloitte Touche Tohmatsu Auditores Independentes. The FCA does not have the dates needed to count the outgoing firm's fiscal years.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 39 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:38:03+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.

Annex: full list of differences in the following year's comparative (R$ million)

(a) DFC (indirect method) 6.01 Caixa Líquido Atividades Operacionais: 45.3 in the version used, 78.6 in the comparative (difference of 33.3)
(a) BPA 1 Ativo Total: 1,126.7 in the version used, 1,114.3 in the comparative (difference of −12.3)
(a) BPP 2.02 Passivo Não Circulante: 335.9 in the version used, 323.5 in the comparative (difference of −12.3)
(b) DFC (indirect method) 6.03.03 Empréstimos Tomados: 322.0 in the version used, −126.5 in the comparative (difference of −448.5)
(b) DFC (indirect method) 6.03.02 Pagamentos Juros s/ Capital Próprio e Dividendos: −15.8 in the version used, 322.0 in the comparative (difference of 337.8)
(b) DFC (indirect method) 6.03.06 Reestruturação Societária: −169.4 in the version used, 0.0 in the comparative (difference of 169.4)
(b) DFC (indirect method) 6.03.05 Ações em Tesouraria: −32.9 in the version used, −169.4 in the comparative (difference of −136.5)
(b) DFC (indirect method) 6.03.04 Pagamentos de Empréstimos: −126.5 in the version used, −32.9

in the comparative (difference of 93.6)
(b) BPA 1.01.01 Caixa e Equivalentes de Caixa: 74.8 in the version used, 162.2 in the comparative (difference of 87.5)
(b) BPA 1.01.02 Aplicações Financeiras: 87.5 in the version used, 0.0 in the comparative (difference of −87.5)
(b) BPA 1.01.02.02 Aplicações Financeiras Avaliadas ao Custo Amortizado: 87.5 in the version used, 0.0 in the comparative (difference of −87.5)
(b) BPA 1.01.02.02.01 Títulos Mantidos até o Vencimento: 87.5 in the version used, 0.0 in the comparative (difference of −87.5)
(b) BPP 2.01.05.02.07 Outras Obrigações: 40.3 in the version used, 0.0 in the comparative (difference of −40.3)
(b) DFC (indirect method) 6.01.01.07 Provisão de Juros s/ Empréstimos e Financiamentos: 38.9 in the version used, −0.9 in the comparative (difference of −39.8)
(b) DFC (indirect method) 6.01.01.10 Variação Cambial Líquida de Investimento Exterior: 0.0 in the version used, 38.9 in the comparative (difference of 38.9)
(b) DFC (indirect method) 6.03 Caixa Líquido Atividades de Financiamento: −23.4 in the version used, −56.7 in the comparative (difference of −33.3)
(b) DFC (indirect method) 6.03.10 Débitos com Empresas Ligadas: 0.0 in the version used, −33.3 in the comparative (difference of −33.3)
(b) DFC (indirect method) 6.01.01.05 Imp. Renda e Contr. Social Diferidos: −33.5 in the version used, −1.9 in the comparative (difference of 31.6)
(b) DFC (indirect method) 6.01.03 Outros: −28.9 in the version used, −6.1 in the comparative (difference of 22.9)
(b) DFC (indirect method) 6.01.01.01 Lucro Líquido do Exercício: 37.3 in the version used, 17.6 in the comparative (difference of −19.7)
(b) DFC (indirect method) 6.03.01 Participação dos Minoritários: 0.0 in the version used, −15.8 in the comparative (difference of −15.8)
(b) DFC (indirect method) 6.01.01 Caixa Gerado nas Operações: 74.9 in the version used, 88.7 in the comparative (difference of 13.9)
(b) BPA 1.02 Ativo Não Circulante: 376.6 in the version used, 364.3 in the comparative (difference of −12.3)
(b) BPA 1.02.01 Ativo Realizável a Longo Prazo: 62.9 in the version used, 50.6 in the comparative (difference of −12.3)
(b) BPA 1.02.01.06 Tributos Diferidos: 56.1 in the version used, 43.8 in the comparative (difference of −12.3)
(b) BPA 1.02.01.06.01 Imposto de Renda e Contribuição Social Diferidos: 56.1 in the version used, 43.8 in the comparative (difference of −12.3)
(b) BPP 2 Passivo Total: 1,126.7 in the version used, 1,114.3 in the comparative (difference of −12.3)
(b) BPP 2.02.03 Tributos Diferidos: 18.4 in the version used, 6.1 in the comparative (difference of −12.3)
(b) BPP 2.02.03.01 Imposto de Renda e Contribuição Social Diferidos: 18.4 in the version used, 6.1 in the comparative (difference of −12.3)
(b) DFC (indirect method) 6.01.01.06 Equivalência Patrimonial: −1.9 in the version used, 3.7 in the comparative (difference of 5.6)
(b) DRE 3.04.04 Outras Receitas Operacionais: 0.0 in the version used, 4.3 in the comparative (difference of 4.3)
(b) DRE 3.04.05 Outras Despesas Operacionais: −9.0 in the version used, −13.3 in the comparative (difference of −4.3)
(b) DFC (indirect method) 6.01.01.08 Provisão Instrumentos Financeiros Derivativos: 3.7 in the version used, 0.0 in the comparative (difference of −3.7)
(b) DFC (indirect method) 6.01.02 Variações nos Ativos e Passivos: −0.6 in the version used, −4.0 in the comparative (difference of −3.5)
(b) DFC (indirect method) 6.01.02.05 Aumento (Redução) em Contas a Pagar e Provisões: 55.0 in the version used, 51.5 in the comparative (difference of −3.5)
(b) BPP 2.01.06.01.01 Provisões Fiscais: 1.5 in the version used, 0.0 in the comparative

(difference of −1.5)
(b) BPP 2.01.06.01.02 Provisões Previdenciárias e Trabalhistas: 0.0 in the version used, 1.5 in the comparative (difference of 1.5)
(b) DFC (indirect method) 6.01.01.04 Provisão Créditos Liquidação Duvidosa: −0.9 in the version used, 0.0 in the comparative (difference of 0.9)
(b) DFC (indirect method) 6.03.11 Impostos e Taxas a Recolher: 0.0 in the version used, −0.8 in the comparative (difference of −0.8)
(b) BPP 2.01.05.02.05 Saques Cambiais: 39.4 in the version used, 39.6 in the comparative (difference of 0.2)